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Paper Code : BS-M301/BSC 301/BSC301 Mathematics-III (Differential Calculus)

UPID : 003445

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) Find f_x and f_y where $f(x,y) = x^2 + xy$.
- (ii) The value of $\int_1^0 \int_0^1 (x+y) dx dy$ is.....
- (iii) The general solution of $px + \frac{a}{p}$, where $p = \frac{dy}{dx}$ is.....
- (iv) The value of $\frac{1}{D}(x^2)$ is.....
- (v) If a path is considered as a sub graph, then the degree of the intermediate vertices is.....
- (vi) The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if.....
- (vii) If $u + v = x, uv = y$, then $\frac{\partial(x,y)}{\partial(u,v)} =$
- (viii) The value of $\int_0^{\pi/2} \int_0^1 r \sin \theta dr d\theta$ is.....
- (ix) The singular solution of the equation $y = px + a\sqrt{1+p^2}$ is
- (x) The value of $D(\sin^3 x)$ is.....
- (xi) A binary tree has exactly
- (xii) The co-efficient of x^3 in the infinite series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{2}$ is

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- ✓ 2. Show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2} u$ if $u = \log(x^3 + y^3 + z^3 - 3xyz)$ [5]
- ✓ 3. Evaluate $\int_R xy(x+y) dx dy$ over the region R bounded by $y = x^2$ and $y = x$. [5]
- ✓ 4. Solve the ODE: $xp^2 + (y-x)p - y = 0$, where $p \equiv \frac{dy}{dx}$ [5]
5. Find the Taylor's series expansion of $\sin x$. <https://www.makaut.com> [5]
6. Solve by variation of parameters, $\frac{d^2 y}{dx^2} + a^2 y = \cos ax$. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- ✓ 7. (a) Find the maxima and minima of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$. Also find the saddle points. [10]
- (b) Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$ [5]

8. (a) Evaluate $\iint y dx dy$ over part of the plane bounded by $y = x$ and the parabola $y = 4x - x^2$. [8]
 (b) Change the order of integration and hence evaluate the integral $\int_0^1 \int_e^e \frac{dx dy}{y^2 \log y}$. [7]
9. (a) Solve the ODE: $(2x + 3y + 7)dx + (3x - 5y + 2)dy = 0$. [8]
 (b) Find the orthogonal trajectory of the family of curve $x^2 + y^2 = r^2$, where r being the variable of parameter. [7]
10. Solve the ODE: $(x^2 D^2 + 3x D + 2)y = \cos(\log x)$, where $D \equiv \frac{d}{dx}$ [15]
11. Describe Kruskal and Prim's algorithm for finding the minimal spanning tree with examples. [15]

*** END OF PAPER ***